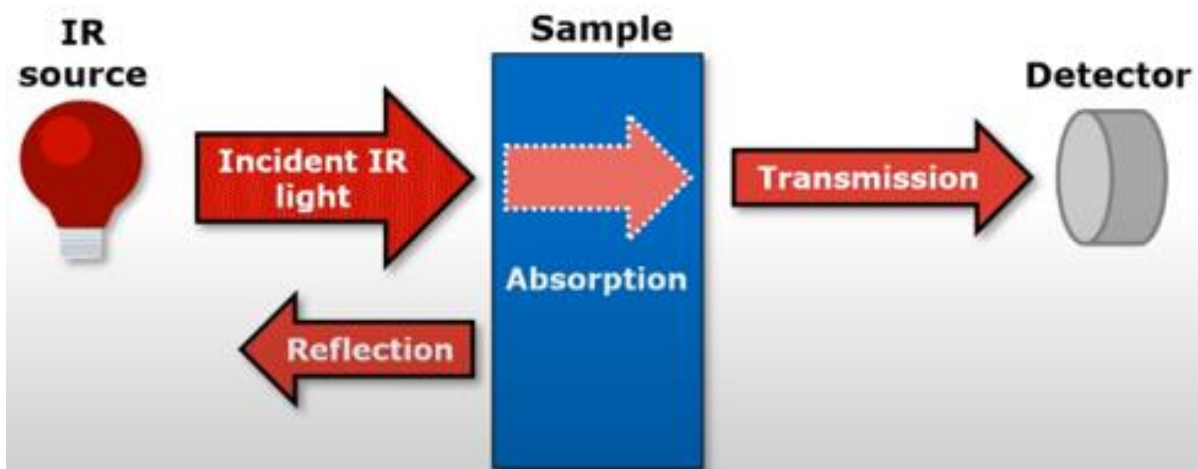
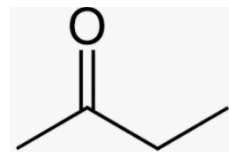
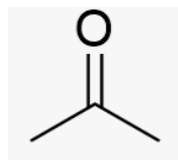


Infrared spectroscopy

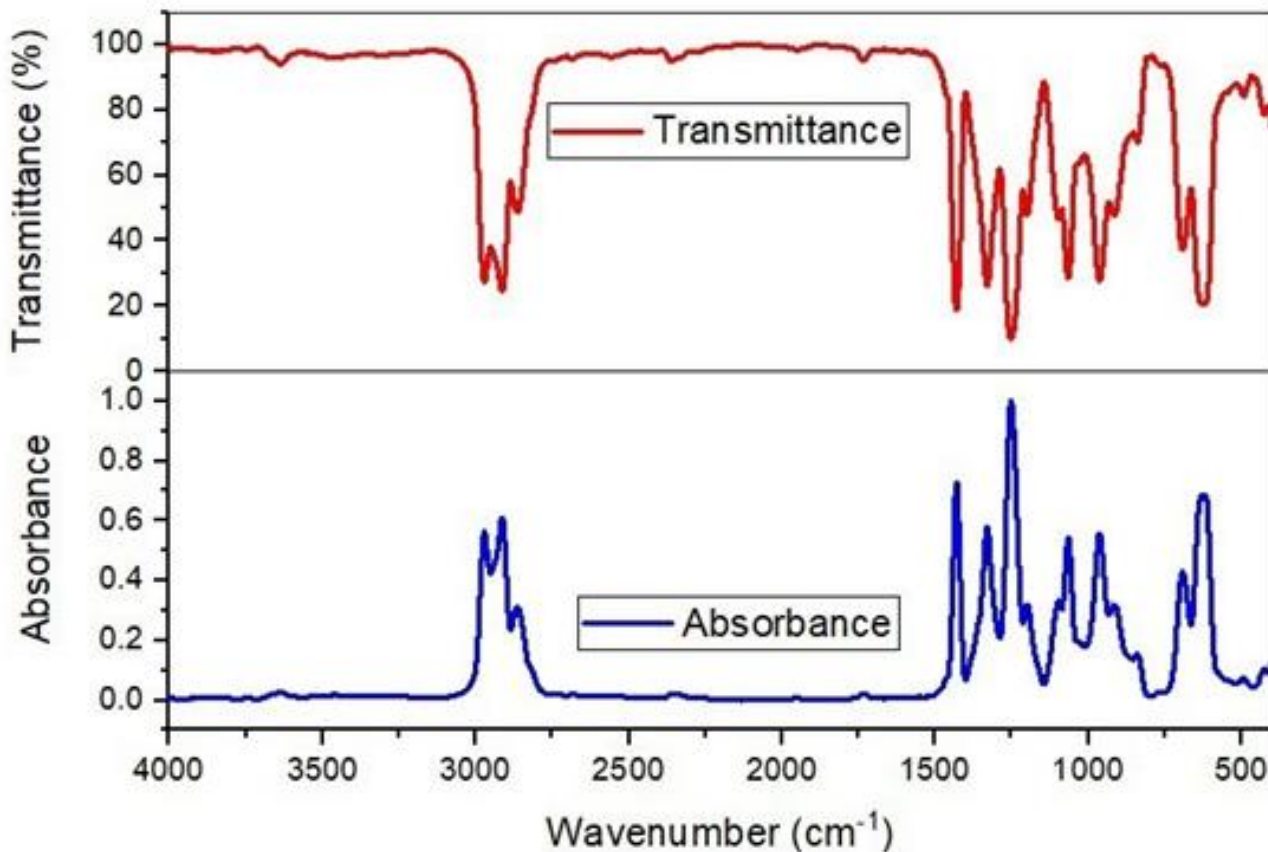
- Mostly for qualitative analysis
- Absorption spectra is recorded as transmittance spectra



- Absorption occurs at specific wavelengths
- IR spectra is called “fingerprints” (600-1400 cm^{-1})
- No other chemical species will have identical IR spectrum (UV can be identical)



Comparison between transmittance (upper) vs absorbance (lower)



The transmittance spectra provide better contrast between intensities of strong and weak bands compared to absorbance spectra.

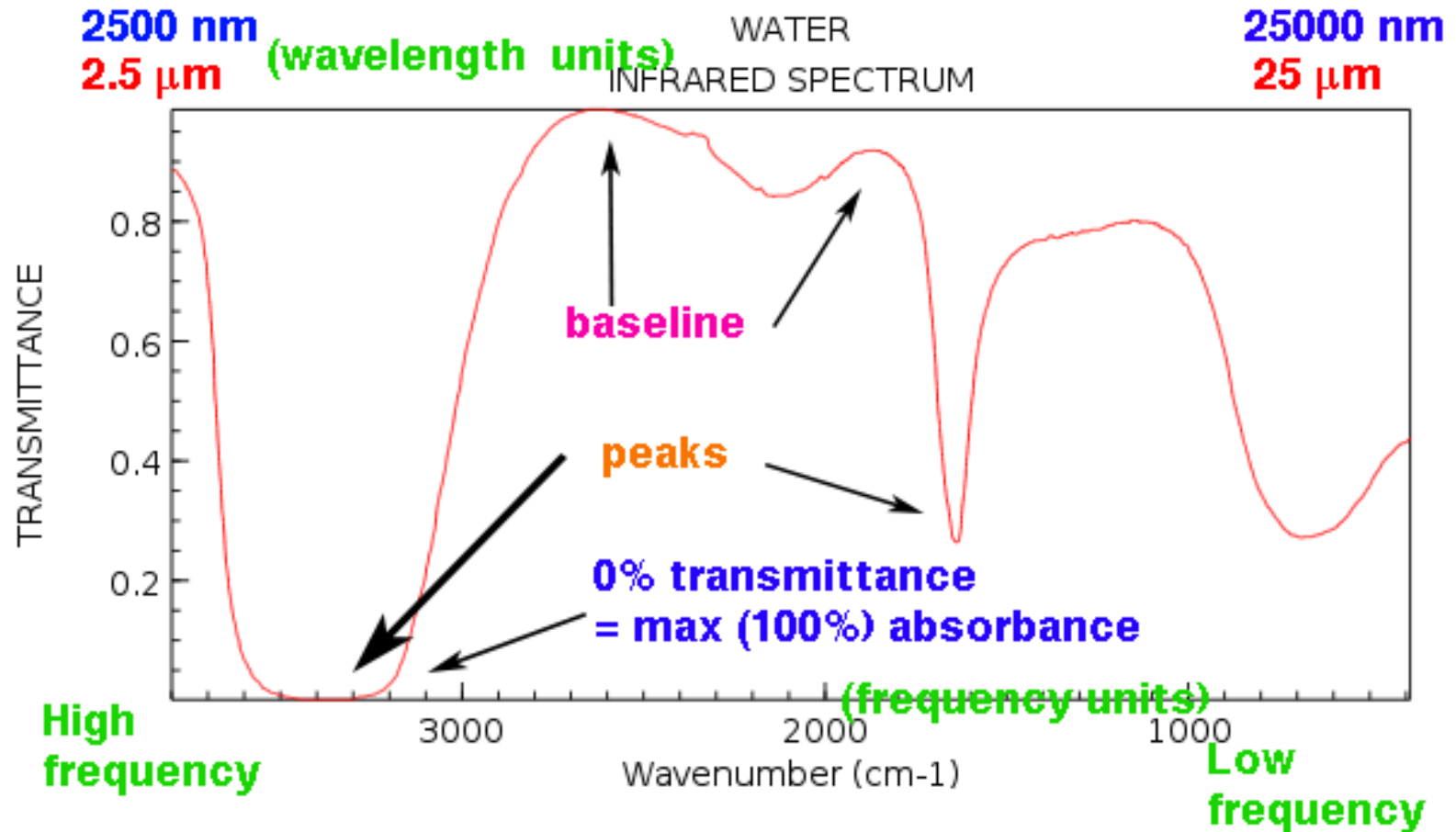
One reason: you identify weak transitions more easily by looking at transmission (more light transmitted than absorbed), and if you're normalizing spectral intensities, those weak ones get washed out if you went by absorption.

What does a higher transmittance mean in IR?

Thus, a high percent transmittance indicates that **little to no light is being absorbed**. The opposite is true if the sample has a low percent transmittance. A low percent transmittance indicates the greater amount of light absorbed; if a sample absorbs light fully, **$i = 0$, and percent transmittance = 0**

The formula for percent transmission is **percent transmission = $i/i_0 \times 100\%$** . “ i ” stands for the intensity of light leaving the sample, and i_0 stands for the intensity of light entering the sample. If the sample does not absorb light, i is equivalent to i_0 , and thus the percent transmittance = **100%**. Thus, a high percent transmittance indicates that little to no light is being absorbed. The opposite is true if the sample has a low percent transmittance.

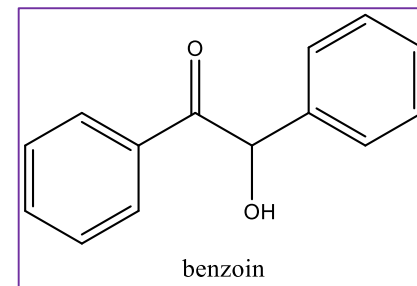
Infrared Spectroscopy



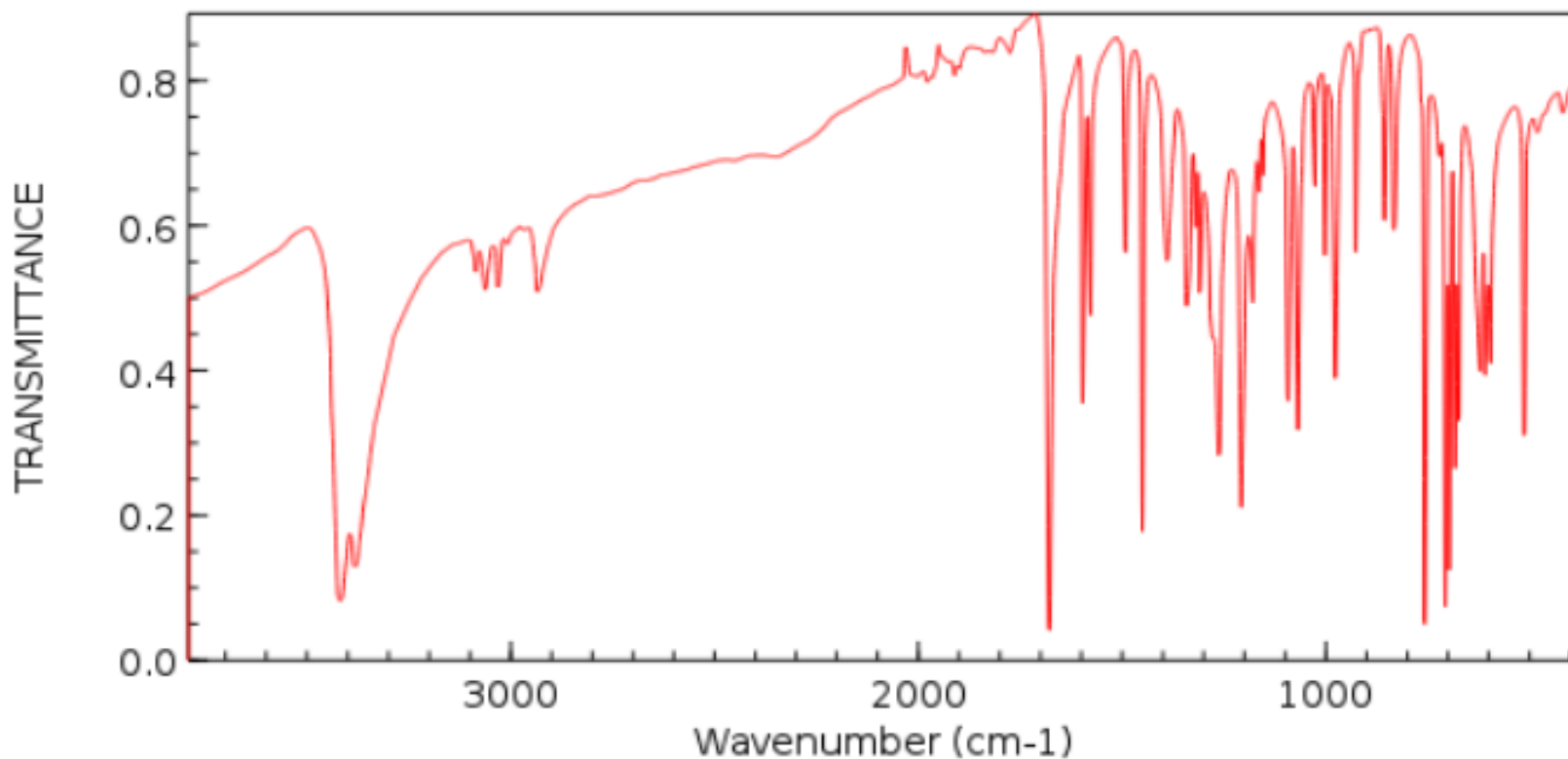
Interpretation of Infrared Spectra

- IR spectrum is due to specific structural features, **a specific bond, within the molecule**, since the vibrational states of individual bonds represent vibrational transition.
- e.g. IR spectrum can tell the molecule has an **O-H bond or a C=O or an aromatic ring**.
- UV-Vis can not tell the presence of O-H but IR can.

Infrared Spectroscopy



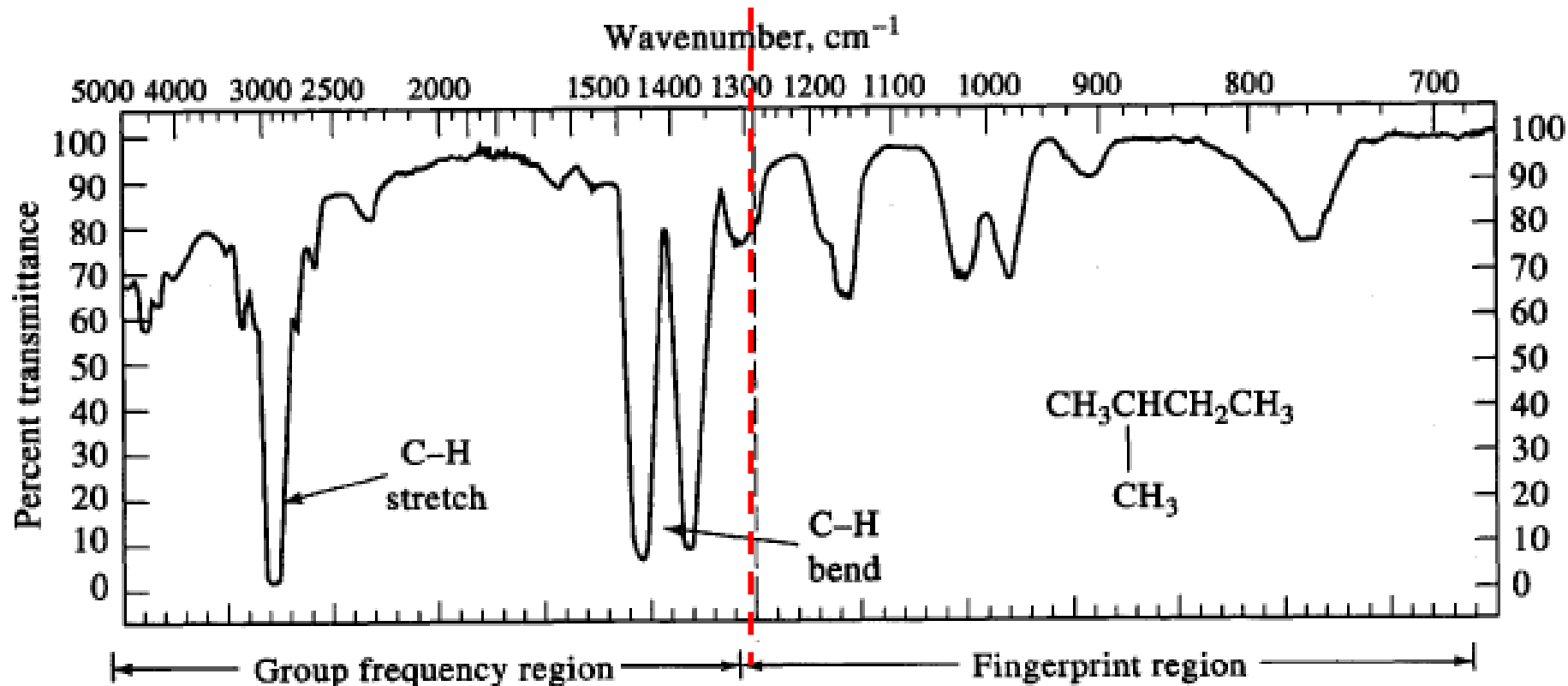
BENZOIN
INFRARED SPECTRUM



Fingerprint of Molecule

- Whole-molecule vibrations and bending vibrations are also quantitized.
- No two molecules will give exactly the same IR spectrum (except enantiomers).
- Simple stretching: $1400\text{-}4000\text{ cm}^{-1}$, “functional group region.”
- Complex vibrations: $600\text{-}1400\text{ cm}^{-1}$, called the “fingerprint region.”

Infrared spectroscopy

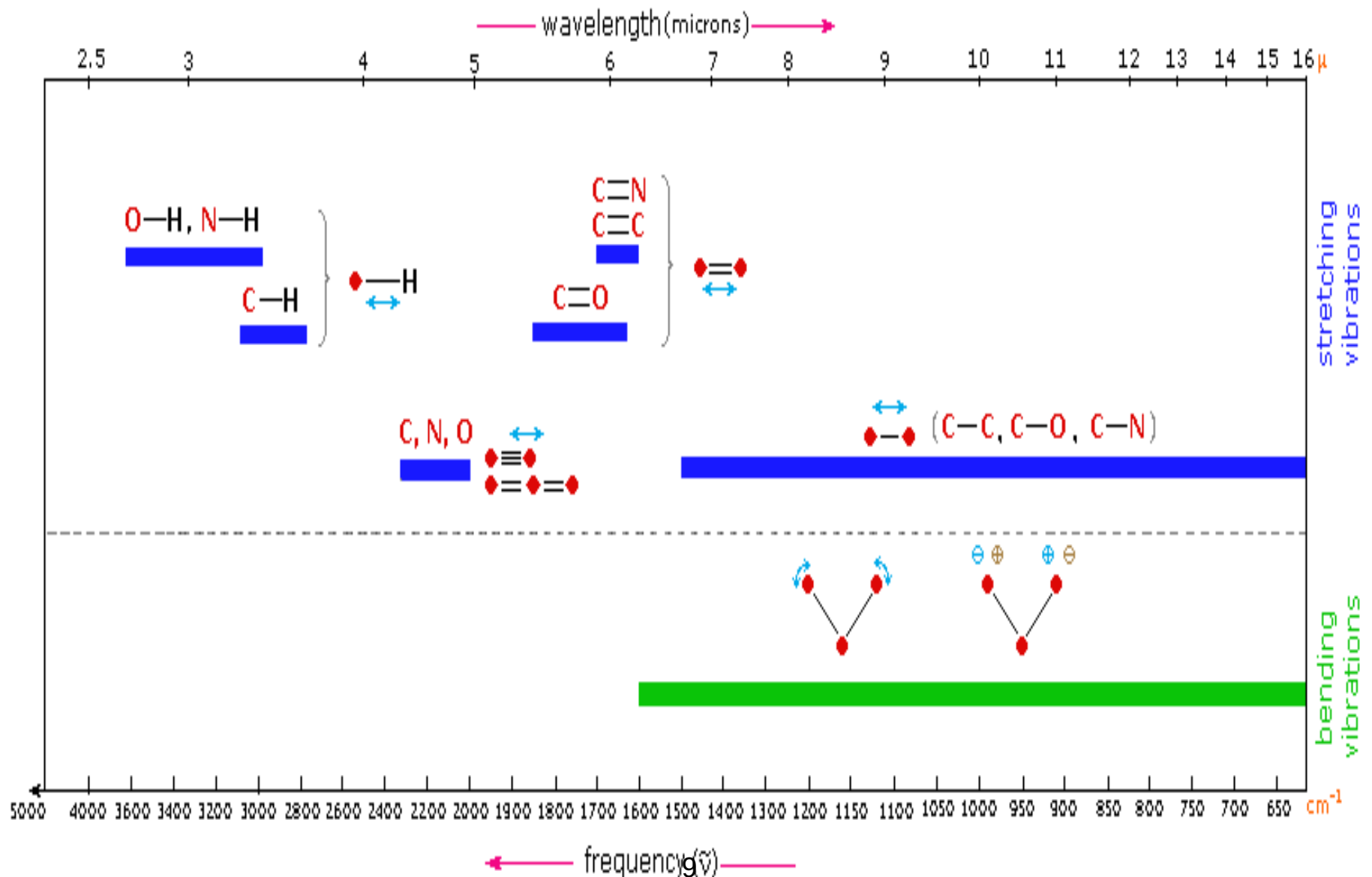


Qualitative Analysis:

Step I: Identify the **functional groups** (group frequency region)

Step II: Compare with standard spectra containing these functional groups
(**fingerprint region**)

Infrared Spectroscopy



How to analyze IR spectra

1. Begin by looking in the region from 4000-1300 cm^{-1} .

Look at the C–H stretching bands around **3000**:

Are any or all to the **right** of 3000
(**less than 3000**)?

Are any or all to the **left** of 3000
(**higher than 3000**)?

Indicates:

alkyl groups (present in most organic molecules)

a C=C bond or aromatic group in the molecule

2. Look for a carbonyl in the region 1760-1690.

If there is such a band:

Is an O–H band also present?

Is a C–O band also present?

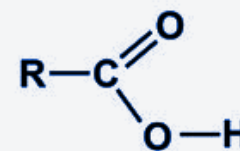
Is an aldehyde C–H band also present?

Is an N–H band also present?

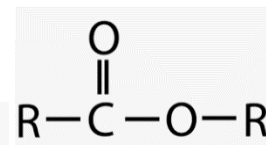
Are none of the above present?

Indicates:

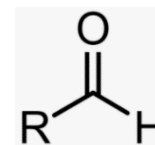
a carboxylic acid group



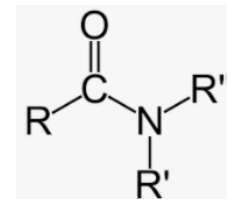
an ester



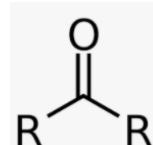
an aldehyde



an amide



a ketone



(also check the [exact position of the carbonyl band](#) for clues as to the type of carbonyl compound it is)

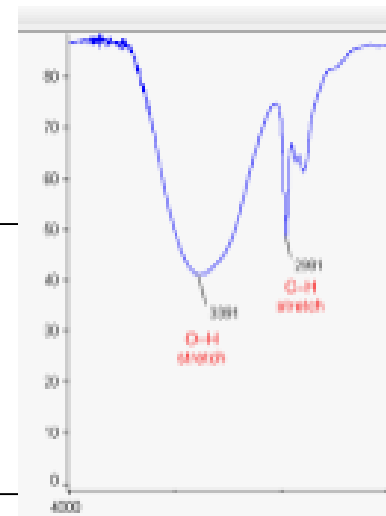
3. Look for a broad O–H band in the region $3500\text{--}3200\text{ cm}^{-1}$.

If there is such a band:

Is an O–H band present?

Indicates:

an alcohol or phenol



4. Look for a single or double sharp N–H band in the region $3400\text{--}3250\text{ cm}^{-1}$.

If there is such a band:

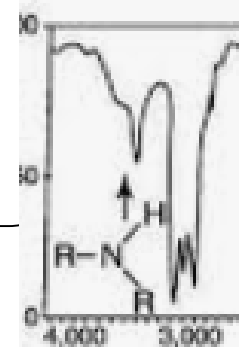
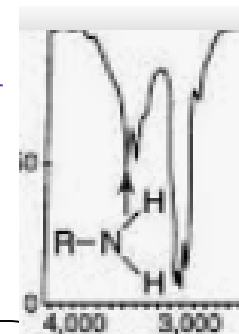
Are there two bands?

Indicates:

a primary amine

Is there only one band?

a secondary amine



5. Other structural features to check for:

Indicates:

Are there C–O stretches?

an ether (or an ester if there is a carbonyl band too)

Is there a C=C stretching band?

an alkene

Are there aromatic stretching bands?

an aromatic

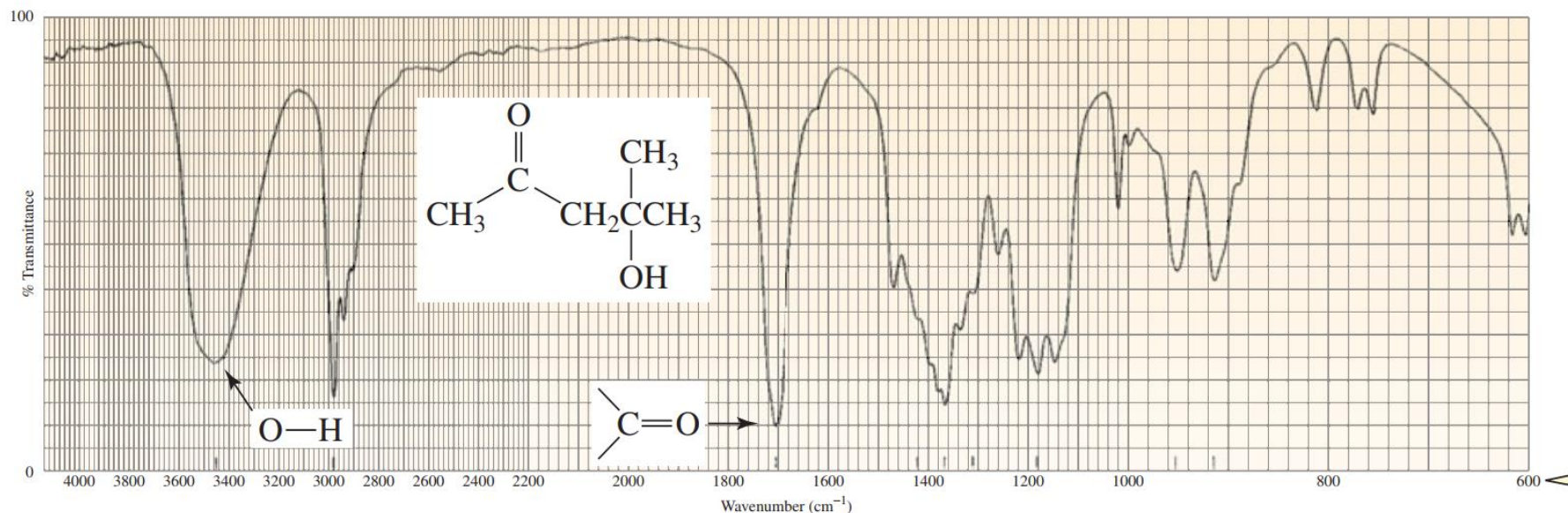
Is there a C≡C band?

an alkyne

Are there -NO₂ bands?

a nitro compound

The stretching vibration of a C=O bond absorbs energy of wave number
 $\sim 1700\text{ cm}^{-1}$ where as the stretching vibration of an O-H bond absorbs energy
of wave number $\sim 3550\text{ cm}^{-1}$

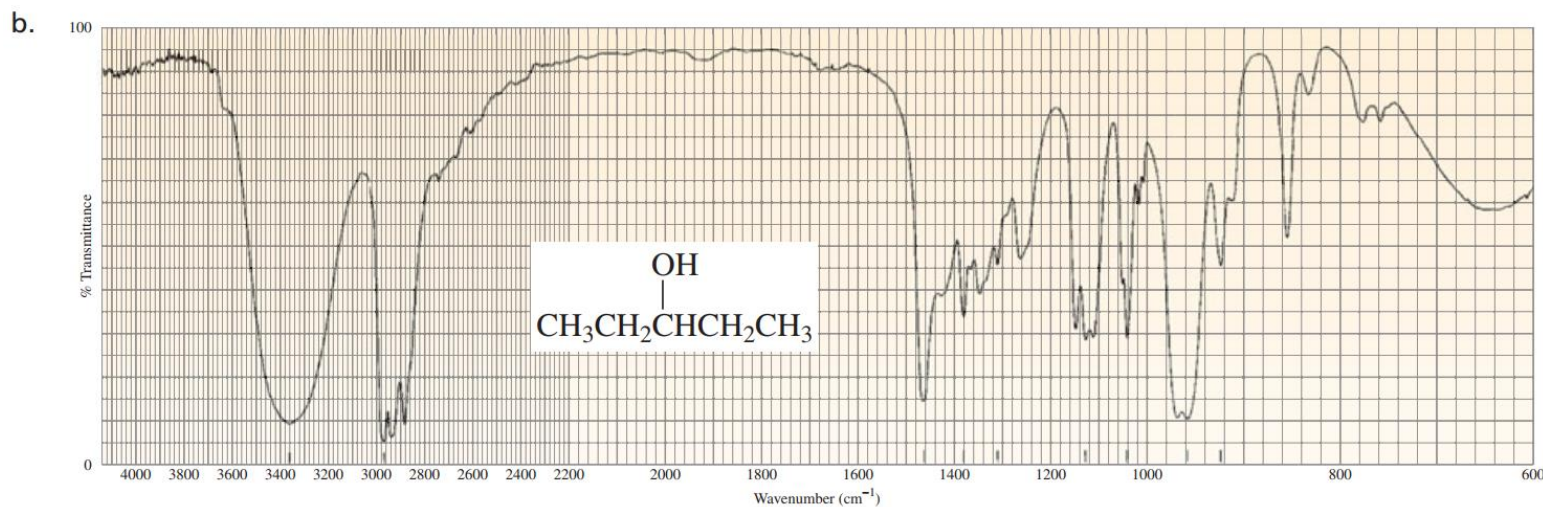
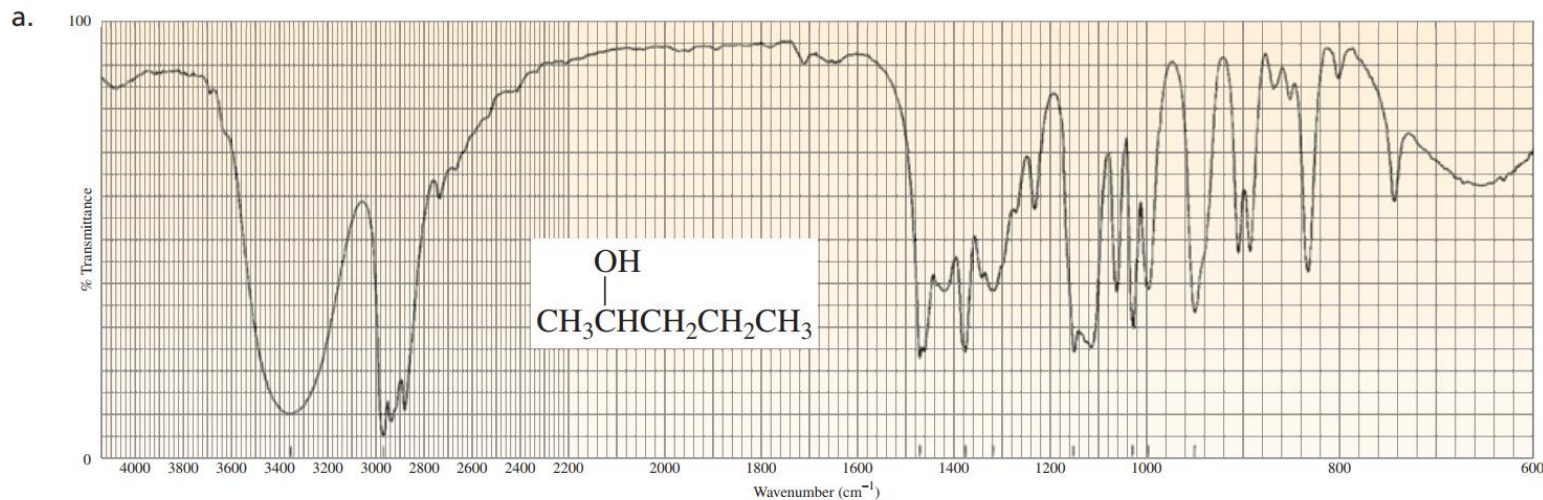


An infrared spectrum shows the percent transmission of radiation versus the wavenumber of the radiation. The (C=O) stretch absorbs at 1705 cm^{-1} and the (O—H) stretch absorbs at 3450 cm^{-1} .

The Functional Group and Fingerprint Regions:.

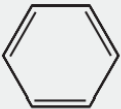
Functional group region: $1400\text{--}4000\text{ cm}^{-1}$

Fingerprint region: $600\text{--}1400\text{ cm}^{-1}$



CHARACTERISTIC INFRARED ABSORPTION BANDS

Table 14.4 Frequencies of Important IR Stretching Vibrations

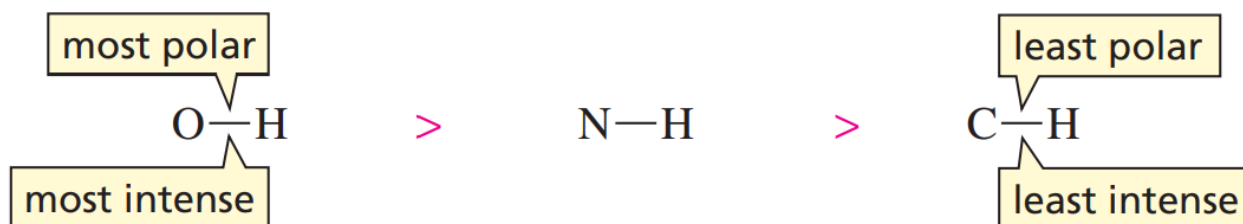
Type of bond	Wavenumber (cm^{-1})	Intensity
$\text{C}\equiv\text{N}$	2260–2220	medium
$\text{C}\equiv\text{C}$	2260–2100	medium to weak
$\text{C}=\text{C}$	1680–1600	medium
$\text{C}=\text{N}$	1650–1550	medium
	~1600 and ~1500–1430	strong to weak
$\text{C}=\text{O}$	1780–1650	strong
$\text{C}-\text{O}$	1250–1050	strong
$\text{C}-\text{N}$	1230–1020	medium
$\text{O}-\text{H}$ (alcohol)	3650–3200	strong, broad
$\text{O}-\text{H}$ (carboxylic acid)	3300–2500	strong, very broad
$\text{N}-\text{H}$	3500–3300	medium, broad
$\text{C}-\text{H}$	3300–2700	medium

THE INTENSITY OF ABSORPTION BANDS

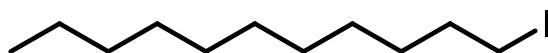
The greater the change in dipole moment, the more intense the absorption

relative bond polarities

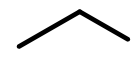
relative intensities of IR absorption



Intensity of absorption depends on the number of bonds responsible for the absorption.



23 C-H bond



5 C-H bond

THE POSITION OF ABSORPTION BANDS

The frequency of a stretching vibration—the amount of energy required to stretch a bond—depends on the *strength* of the bond and the *masses* of the bonded atoms. The stronger the bond, the greater the energy required to stretch it. The frequency of the stretching vibration is also inversely related to the mass of the atoms joined by the bond; thus, heavier atoms vibrate at lower frequencies.

Hooke's Law

The approximate wavenumber of an absorption band can be calculated from the following equation derived from **Hooke's law**, which describes the motion of a vibrating spring:

$$\tilde{\nu} = \frac{1}{2\pi c} \left[\frac{f(m_1 + m_2)}{m_1 m_2} \right]^{1/2}$$

where $\tilde{\nu}$ is the wavenumber of an absorption band, c is the speed of light, f is the force constant of the bond (a measure of the strength of the bond), and m_1 and m_2 are the masses of the atoms (in grams) joined by the bond. This equation shows that *stronger bonds* and *lighter atoms* give rise to higher frequencies.

Lighter atoms shows absorption bands at larger wave number (more energy)

C-H
~3000 cm⁻¹

C-D
~2200 cm⁻¹

C-O
~1100 cm⁻¹

C-Cl
~700 cm⁻¹

Stretching Frequencies

Bond	Bond Energy [kcal (kJ)]	Stretching Frequency (cm ⁻¹)
<i>Frequency dependence on atomic masses</i>		
C—H	100 (420)	3000
C—D	100 (420)	2100
C—C	83 (350)	1200
↓ heavier atoms		↓ $\bar{\nu}$ decreases
<i>Frequency dependence on bond energies</i>		
C—C	83 (350)	1200
C=C	146 (611)	1660
C≡C	200 (840)	2200
↓ stronger bond		↓ $\bar{\nu}$ increases

- Frequency **decreases** with **increasing** atomic weight.
- Frequency **increases** with **increasing** bond energy.

Effect of bond order

- Stronger bonds absorb at higher frequencies:

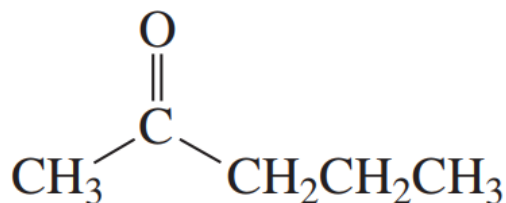
• $\text{C}\equiv\text{C}$ $-\text{C}=\text{C}$ $-\text{C}-\text{C}$
~2200 cm^{-1} ~1660 cm^{-1} ~ 1100 cm^{-1}
(weak or absent if internal)

$-\text{C}=\text{O}$ $-\text{C}-\text{O}$
~1700 cm^{-1} ~1100 cm^{-1}

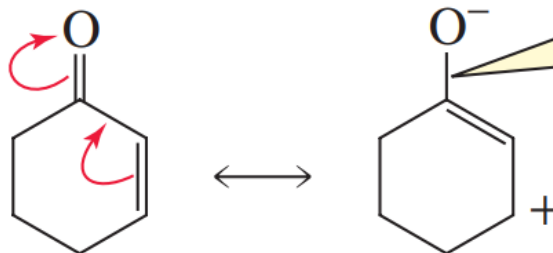
$-\text{C}\equiv\text{N}$ $-\text{C}=\text{C}$ $-\text{C}-\text{C}$
~2200 cm^{-1} ~1600 cm^{-1} ~ 1100 cm^{-1}

Conjugation lowers the frequency

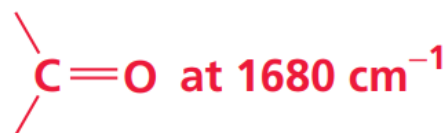
isolated C=C	1640-1680 cm^{-1}
conjugated C=C	1620-1640 cm^{-1}
aromatic C=C	approx. 1600 cm^{-1}



2-pentanone



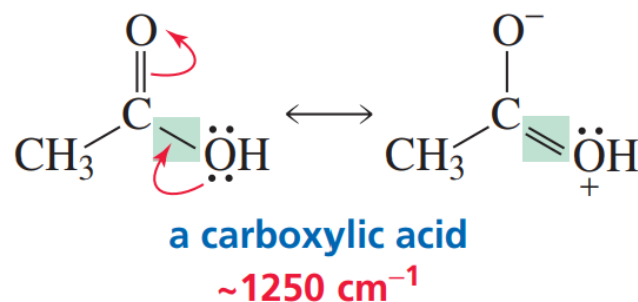
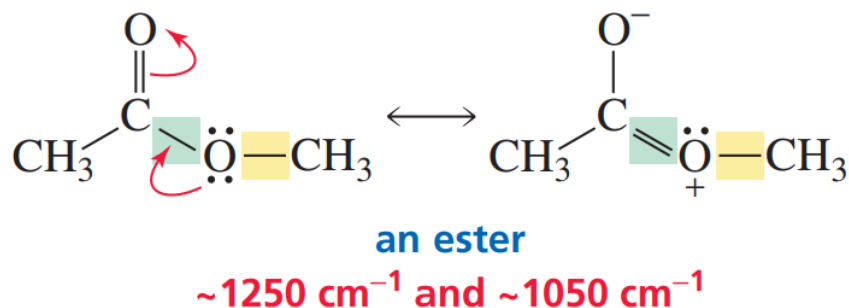
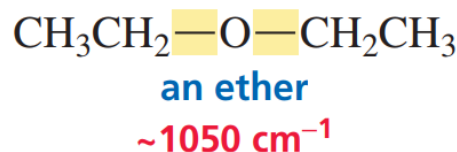
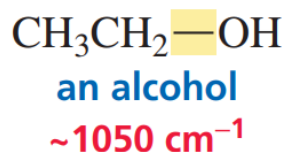
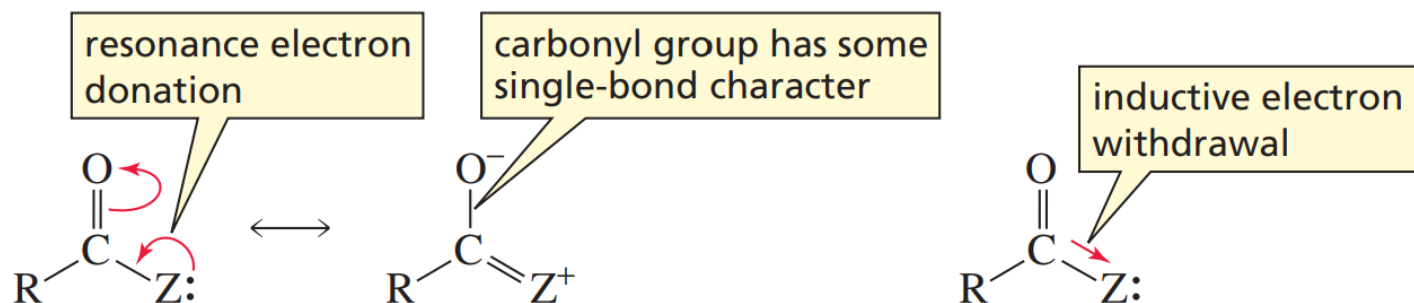
2-cyclohexenone



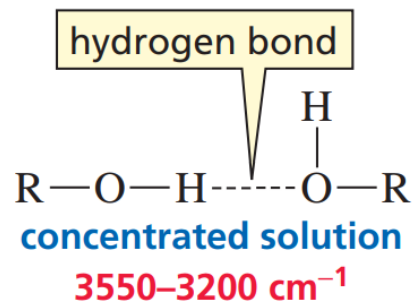
carbonyl group has some single-bond character

Resonance and Inductive effect

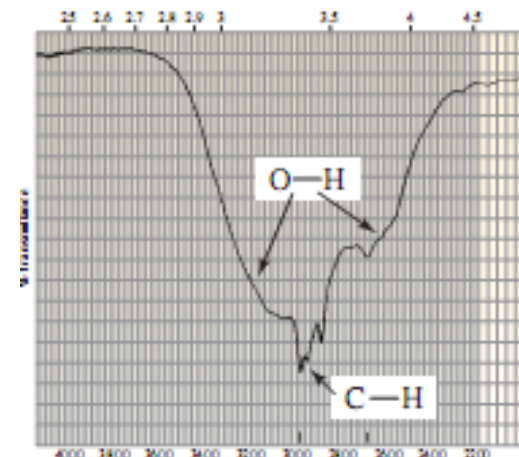
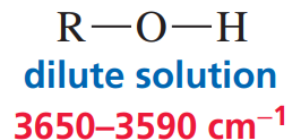
Putting an atom other than carbon next to the carbonyl group also causes the position of the carbonyl absorption band to shift. Whether it shifts to a lower or to a higher frequency depends on whether the predominant effect of the atom is to donate electrons by resonance or to withdraw electrons inductively.



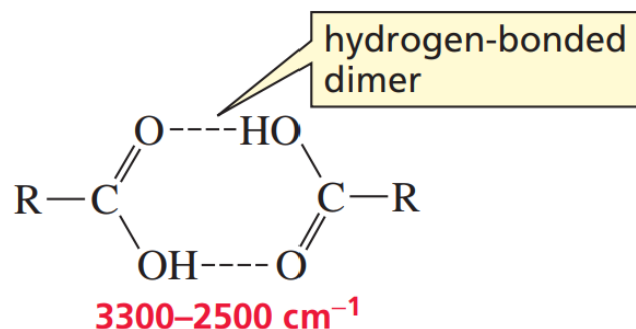
Infrared spectroscopy



Dilute solution with non hydrogen-bonding organic solvents



The shift in frequency of about 300 cm^{-1} arises because hydrogen bonding weakens the O-H bond.



Carboxylic acids exist as **hydrogen-bonded dimers**. The additional hydrogen bonding of carboxylic acids compared with the hydrogen bonding of alcohols causes the O-H stretch of a carboxylic acid to occur at a **lower frequency** and to be broader ($3300-2500 \text{ cm}^{-1}$) than the O-H stretch of an alcohol ($3550-3200 \text{ cm}^{-1}$).

C—H Absorption Bands

The strength of C-H bond depends on the hybridization of the carbon: **The greater the “s” character of the carbon, the stronger the bond it forms. This is opposite to the concept of acidity.**

Table 13.5 IR Absorptions of Carbon-Hydrogen Bonds

Carbon-Hydrogen Stretching Vibrations	Wavenumber (cm ⁻¹)
C≡C—H	~3300
C=C—H	3100–3020
C—C—H	2960–2850
$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{C}-\text{H} \end{array}$	~2820 and ~2720

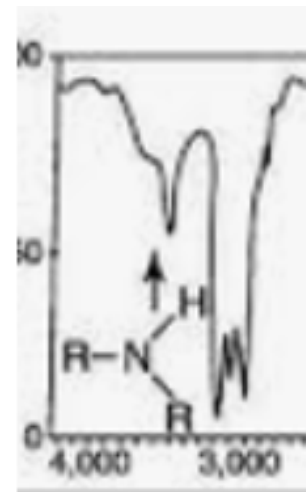
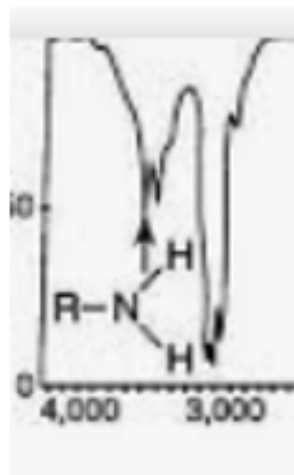
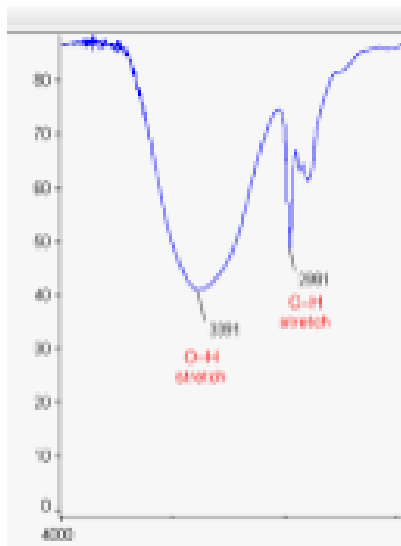
sp^3 C-H, just below 3000 cm⁻¹ (to the right)

sp^2 C-H, just above 3000 cm⁻¹ (to the left)

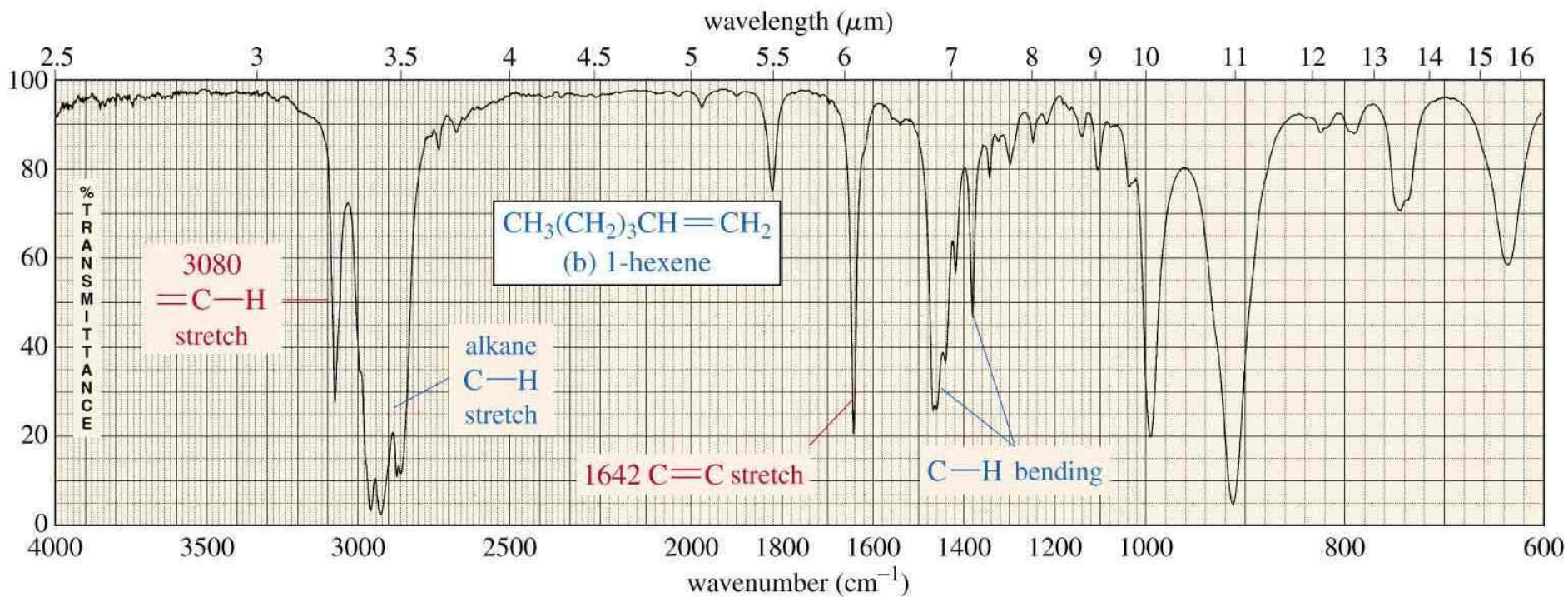
sp C-H, at 3300 cm⁻¹

O-H and N-H Stretching

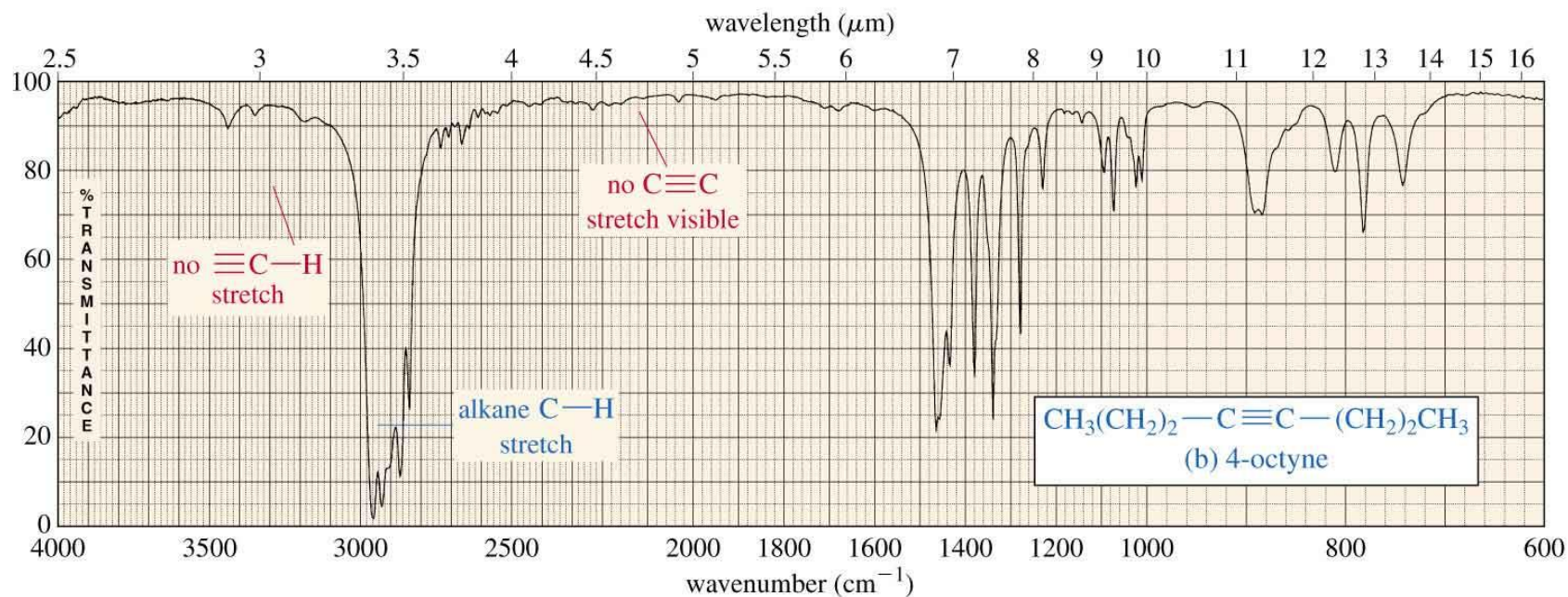
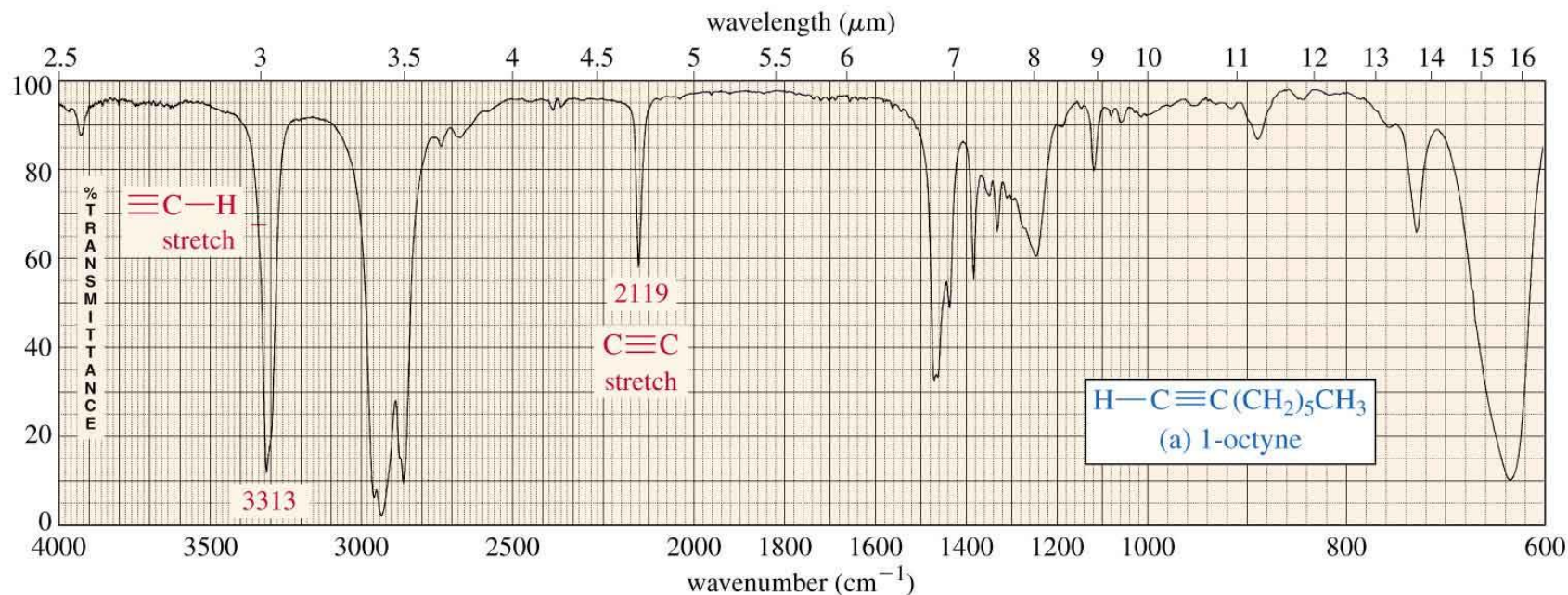
- Both of these occur around 3300 cm^{-1} , but they look different.
 - Alcohol O-H, broad with rounded tip.
 - Secondary amine (R_2NH), broad with one sharp spike.
 - Primary amine (RNH_2), broad with two sharp spikes.
 - No signal for a tertiary amine (R_3N)



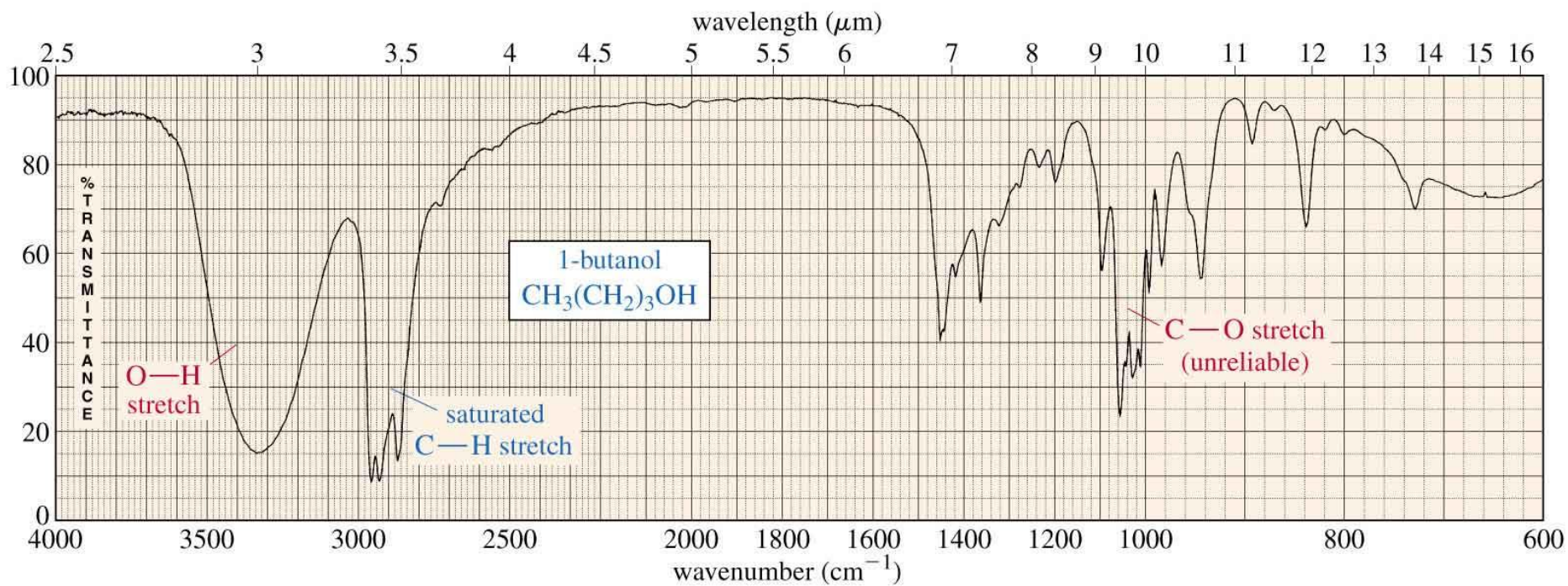
An Alkene IR Spectrum



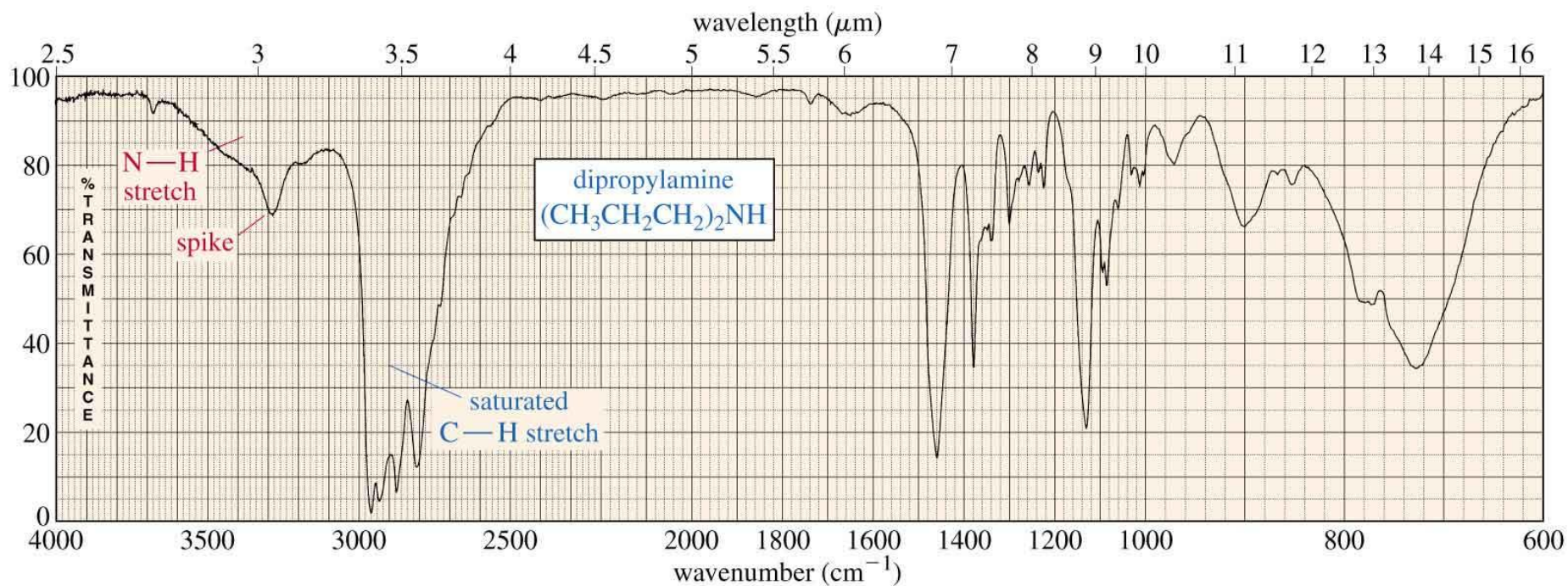
Alkyne IR Spectrum



An Alcohol IR Spectrum



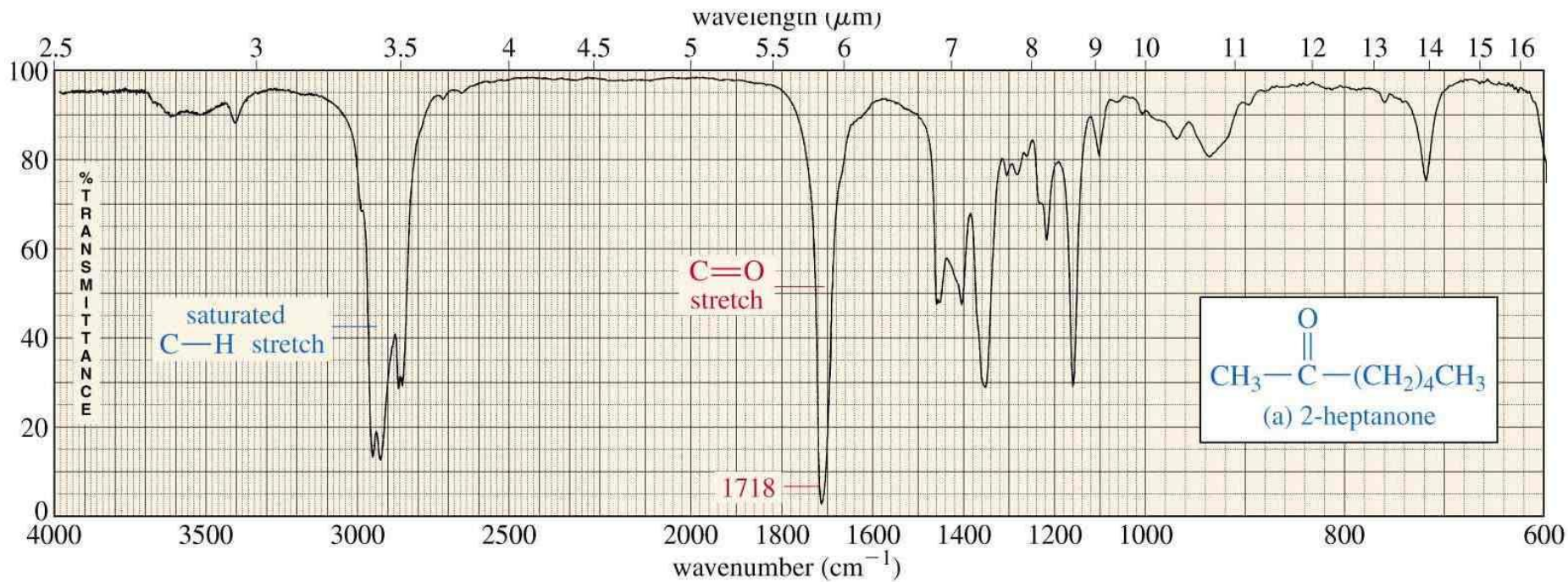
An Amine IR Spectrum



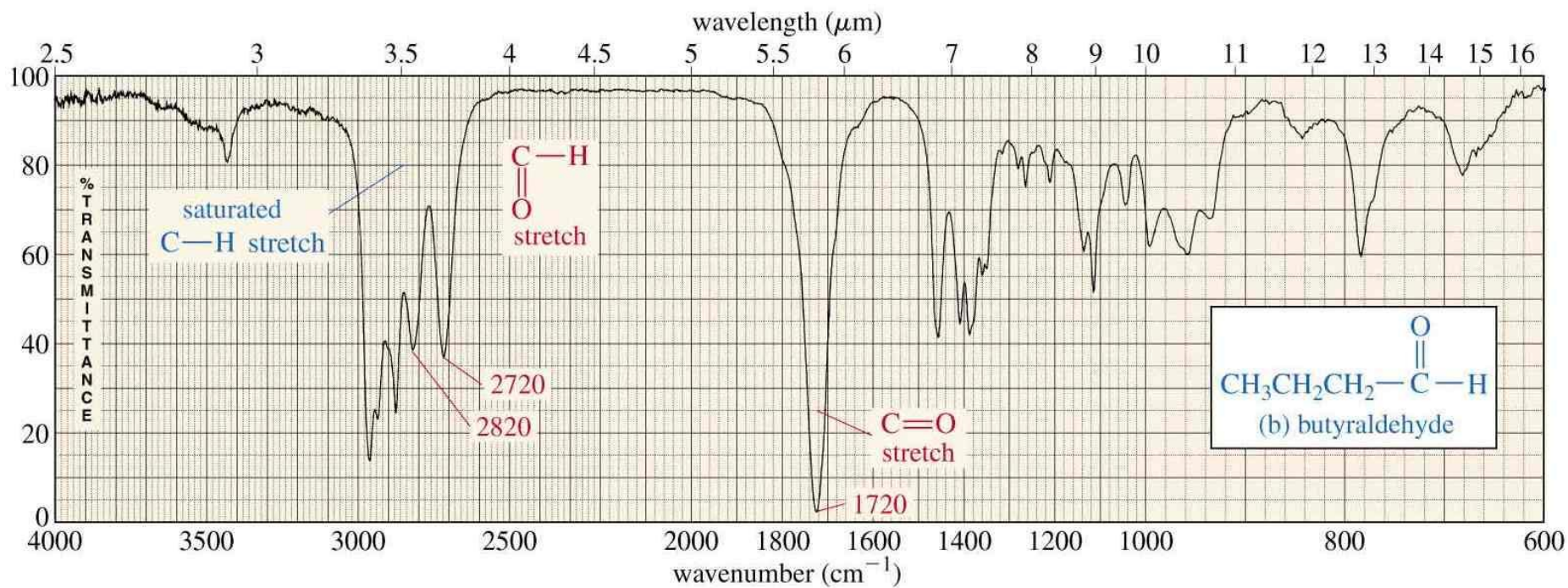
Carbonyl Stretching

- The C=O bond of simple ketones, aldehydes, and carboxylic acids absorb around **1710 cm⁻¹**.
- Usually, it's the **strongest IR** signal.
- Carboxylic acids will have **O-H also**.
- Aldehydes have two C-H signals around **2700** and **2800 cm⁻¹**.

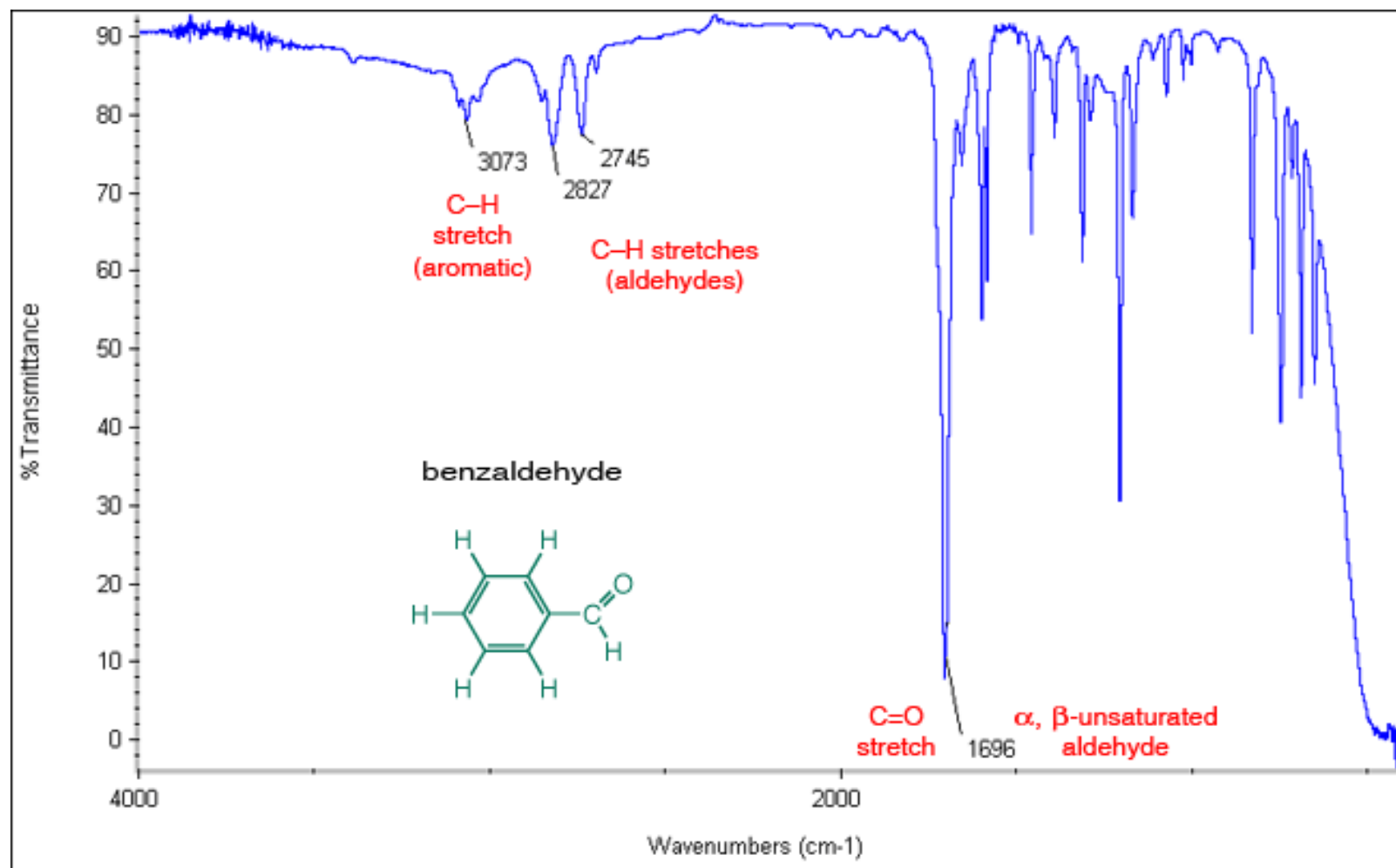
A Ketone IR Spectrum



An Aldehyde IR Spectrum

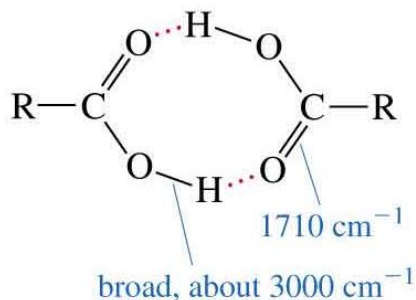
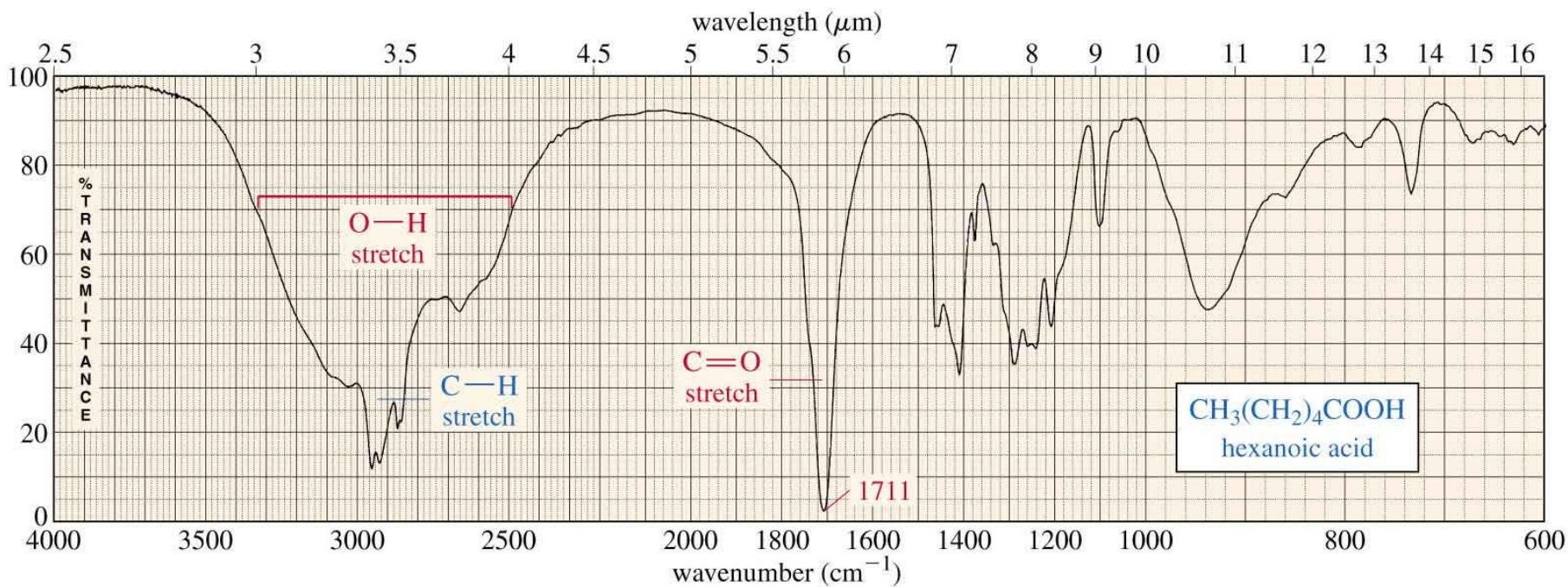


An Aldehyde IR Spectrum

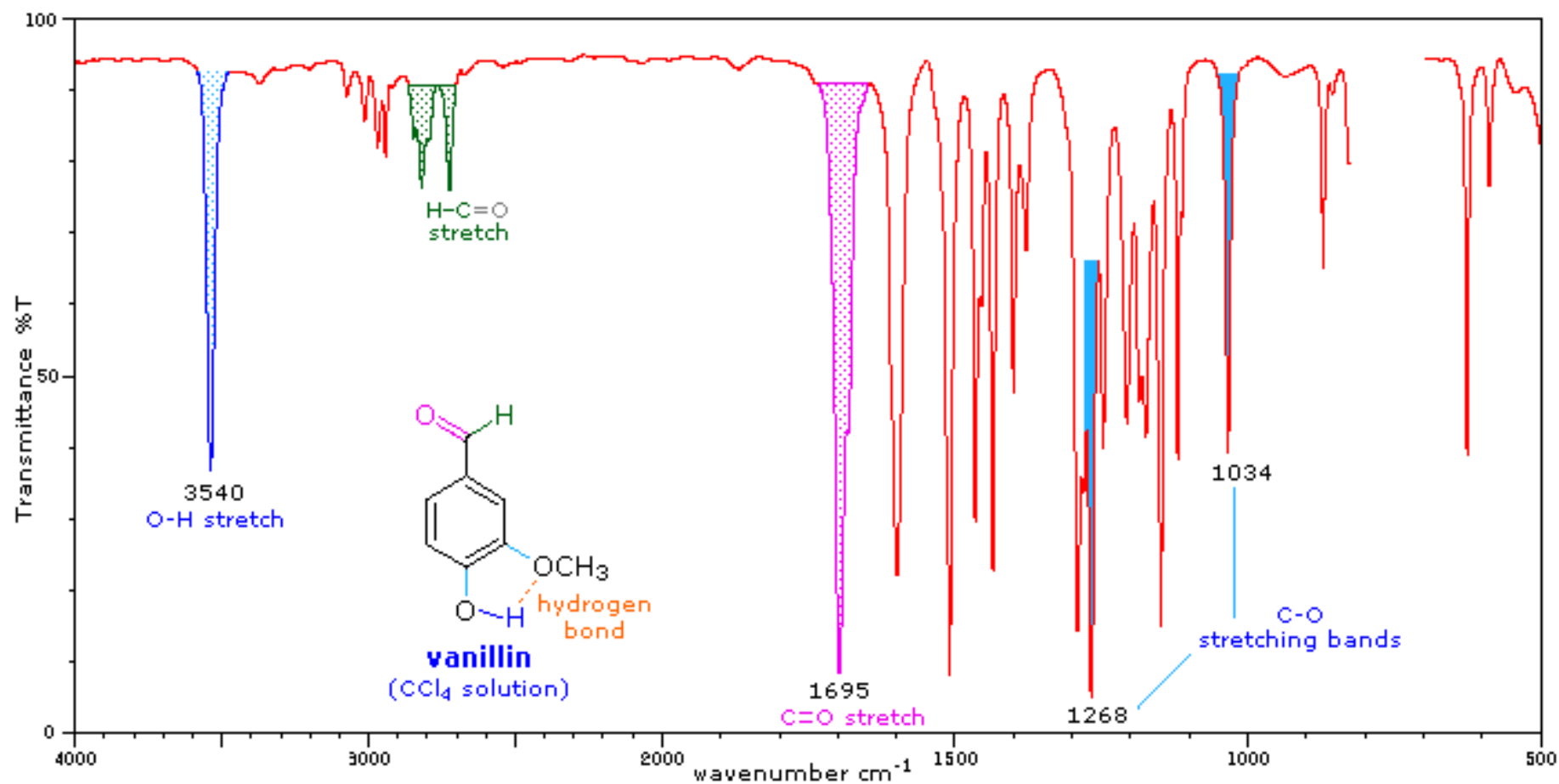


O-H Stretch of a Carboxylic Acid

This O-H absorbs broadly, 2500-3500 cm^{-1} , due to strong hydrogen bonding.



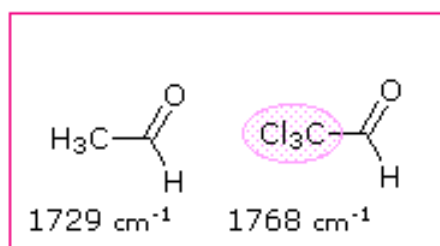
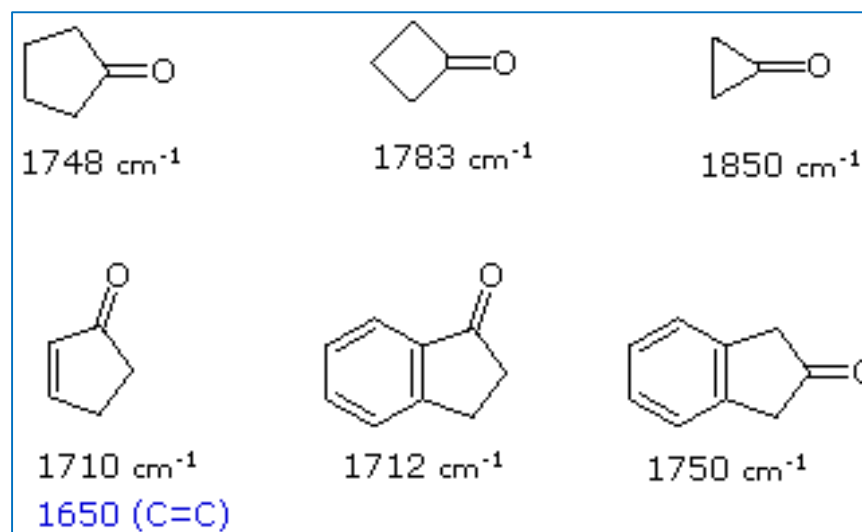
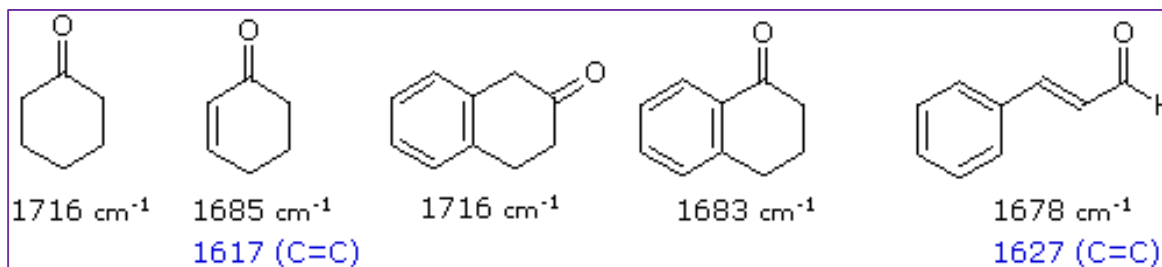
Vanillin IR Spectrum



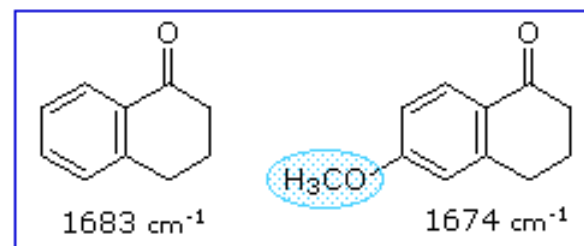
Variations in C=O Absorption

- Conjugation of C=O with C=C lowers the stretching frequency to **$\sim 1680\text{ cm}^{-1}$** .
- The C=O group of an amide absorbs at an even lower frequency, **1640-1680 cm^{-1}** .
- The C=O of an ester absorbs at a higher frequency, **$\sim 1730\text{-}1740\text{ cm}^{-1}$** .
- Carbonyl groups in **small rings** (5 C's or less) absorb at an **even higher frequency (Because of higher s character)**.

Variations in C=O Absorption



electron withdrawing substituent

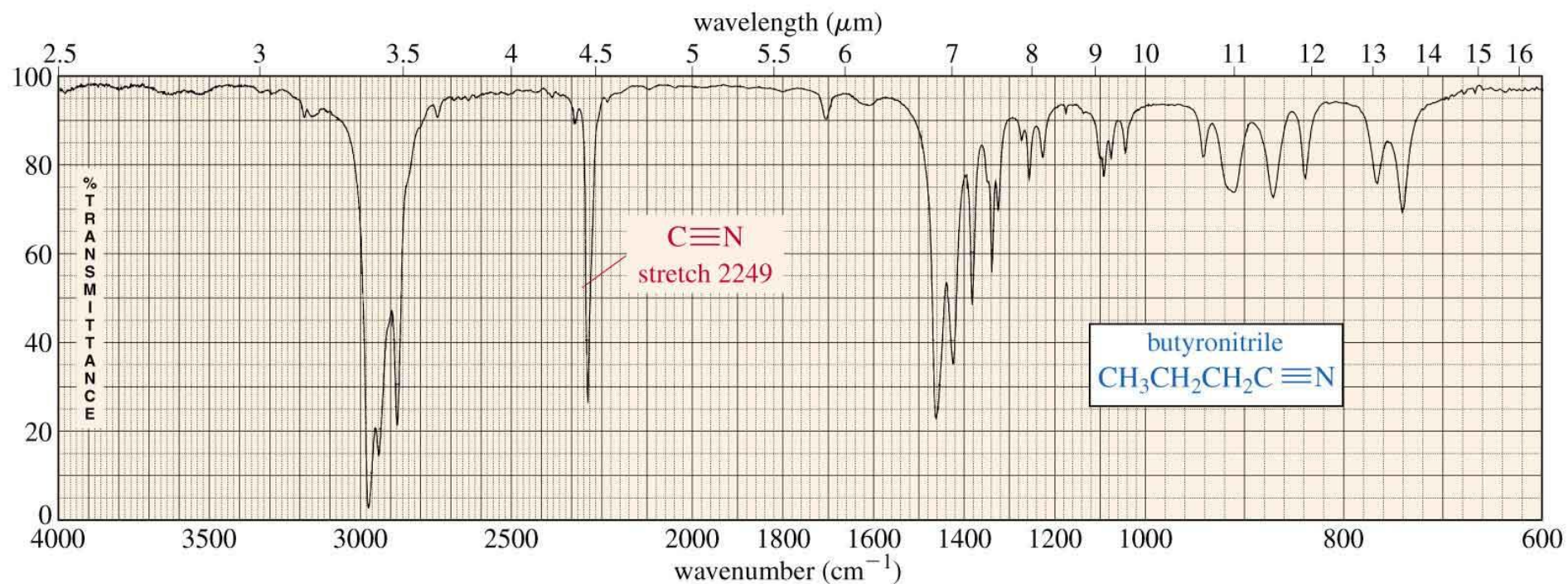


electron donating substituent

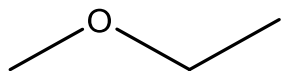
Carbon - Nitrogen Stretching

- C - N absorbs around **1200 cm⁻¹**.
- C = N absorbs around **1660 cm⁻¹** and is much stronger than the C = C absorption in the same region.
- C ≡ N absorbs strongly just *above* **2200 cm⁻¹**.
- The alkyne C ≡ C signal is much weaker and is just *below* **2200 cm⁻¹**.

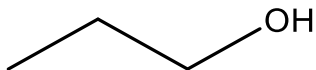
A Nitrile IR Spectrum



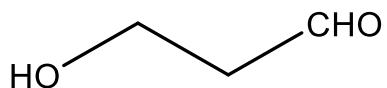
Q. Distinguish between the following pair of compounds.



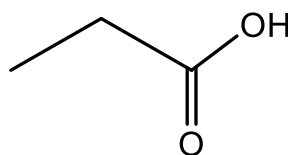
and



Broad -OH stretch at **3300 cm^{-1}**

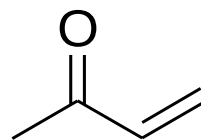
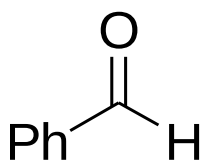
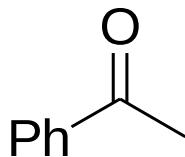
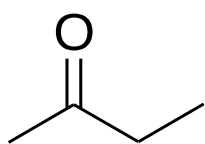


and

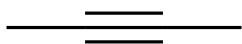


Characteristic **C-H str of aldehyde**

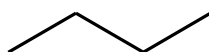
Q . Which of the following compound is expected to show IR absorption at 2710 and 1705.



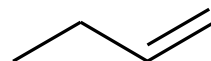
Q . Which of the following compound is expected to show strong IR absorption at 3300.



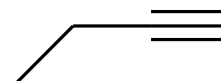
A



B



C



D

Strengths and Limitations

- IR alone **cannot determine a structure**.
- Some signals may be **ambiguous**.
- The functional group **is usually indicated**.
- The ***absence* of a signal is definite proof** that the functional group is **absent**.
- Correspondence with a known sample's IR spectrum **confirms the identity** of the compound.